entailed a radical change in perspective. *Since gravity and acceleration are equivalent, if you feel gravity's influence, you must be accelerating.* Einstein argued that only those observers who feel no force at all—including the force of gravity—are justified in declaring that they are not accelerating. Such force-free observers provide the true reference points for discussing motion, and it's this recognition that requires a major turnabout in the way we usually think about such things. When Barney jumps from his window into the evacuated shaft, we would ordinarily describe him as accelerating down toward the earth's surface. But this is not a description Einstein would agree with. According to Einstein, Barney is *not* accelerating. *He* feels no force. *He* is weightless. *He* feels as he would floating in the deep darkness of empty space. *He* provides the standard against which all motion should be compared. And by this comparison, when you are calmly reading at home, *you* are accelerating. From Barney's perspective as he freely falls by your window—the perspective, according to Einstein, of a true benchmark for motion—you and the earth and all the other things we usually think of as stationary are *accelerating upward*. Einstein would argue that it was Newton's head that rushed up to meet the apple, not the other way around.

Clearly, this is a radically different way of thinking about motion. But it's anchored in the simple recognition that you feel gravity's influence only when you resist it. By contrast, when you fully give in to gravity you don't feel it. Assuming you are not subject to any other influences (such as air resistance), when you give in to gravity and allow yourself to fall freely, you feel as you would if you were freely floating in empty space—a perspective which, unhesitatingly, we consider to be unaccelerated.

In sum, only those individuals who are freely floating, regardless of whether they are in the depths of outer space or on a collision course with the earth's surface, are justified in claiming that they are experiencing no acceleration. If you pass by such an observer and there is relative acceleration between the two of you, then according to Einstein, *you* are accelerating.

As a matter of fact, notice that neither Itchy, nor Scratchy, nor Apu, nor Martin was truly justified in saying that he was stationary during the duel, since they all felt the downward pull of gravity. This has no bearing